

Martin's Axiom, Large Continuum and Global Σ_n^1 -Uniformization

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Abstract

We construct a generic extension of L satisfying Martin's Axiom, $2^{\aleph_0} = \aleph_3$, a lightface Δ_3^1 wellorder of the reals, and global Σ_n^1 -uniformization for every $n \geq 2$. The argument combines the Fischer–Friedman–Zdomskyy branch-versus-specialize coding matrix with an infinite copying construction for projective uniformization. The matrix yields a robust lightface Σ_3^1 predicate $\Psi_3(z)$ whose positive instances are exactly the reals deliberately inserted into uniformization-reserved coding blocks. Since the compression of long real sequences is used only in the external bookkeeping of the iteration, the final projective definitions do not depend on non-definable coding maps. At each projective level $n \geq 3$, absolute Π_2^1 or Σ_2^1 matrix values are copied into Ψ_3 and then used to define Σ_n^1 acceptance predicates selecting the least element of each nonempty vertical section in the final projective wellorder. Together with the Kondo–Novikov theorem at level 2, this gives simultaneous uniformization at all finite projective levels.

Keywords: Martin's Axiom; projective uniformization; projective wellorders; Suslin trees; forcing; cardinal characteristics.

1 Introduction

Uniformization is one of the classical definable choice problems for sets of reals. If $A \subseteq \mathbb{R} \times \mathbb{R}$, a uniformization of A is a function whose graph is contained in A and whose domain is the projection of A on the first coordinate. The projective uniformization problem asks whether, when A belongs to a projective pointclass, such a choice function can be found with a graph of the same projective complexity. The Kondo–Novikov theorem gives the first nontrivial positive result in ZFC: coanalytic sets, and consequently Σ_2^1 sets, admit same-level uniformizations; see, for example, [16]. Above this level the behaviour of uniformization is highly sensitive to the ambient universe.

There are two familiar sources of global information about the projective hierarchy. In L , the existence of a good projective wellorder gives definable choices and hence positive uniformization statements on the Σ -side. Under large-cardinal assumptions leading to projective determinacy, one obtains a very different pattern of regularity, separation, reduction and uniformization. The present paper belongs to a complementary program: the study of separation, reduction and uniformization in non-PD contexts. The aim is to understand

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which fragments of the constructible picture and which fragments of the regularity behaviour suggested by determinacy can be forced from weak hypotheses, and how far these principles can be combined with forcing axioms, definable wellorders, large continuum, and failures of neighbouring projective principles.

This program has developed through several forcing constructions for projective separation, reduction and uniformization. It includes forcing the Σ_3^1 -separation property [6], forcing Π_n^1 -uniformization patterns contrary to the usual PD-guided picture [7], separating Π_3^1 -reduction from Π_3^1 -uniformization [8], and obtaining Σ_3^1 -uniformization from forcing axioms [10]. The global copying construction for Σ -uniformization was developed further in the BPFA setting in [11]. Recent preprints continue the same line by separating reduction from separation, combining projective wellorders with Π -uniformization, mixing upper Σ -uniformization with lower Π -side phenomena, and separating boldface from lightface uniformization at low levels; see [9, 12, 13, 14]. The result below is another instance of this non-PD program.

Theorem 1.1 (Main theorem). *There is a generic extension of L satisfying*

$$\text{MA} + 2^{\aleph_0} = \aleph_3$$

such that the reals have a lightface Δ_3^1 wellorder and, for every $n \geq 2$, every boldface Σ_n^1 set of pairs of reals admits a boldface Σ_n^1 uniformization.

Thus the final universe is rich from the point of view of the projective sets of reals, although no large cardinals are used. It is far from the determinacy universe, since it contains a projective wellorder of the reals. At the same time, Martin's Axiom has genuine regularizing force at the second projective level: in the presence of $\text{MA} + \neg\text{CH}$, every boldface Σ_2^1 set of reals is Lebesgue measurable and has the Baire property; see, for example, [4, 16]. In the model constructed here this low-level measure and category regularity is combined with Kondo–Novikov uniformization at level 2, a lightface Δ_3^1 wellorder, large continuum, and global same-level Σ -uniformization at all finite projective levels. The theorem therefore produces a universe in which the reals are, in several precise senses, quite well behaved, but where this behaviour is obtained by forcing over L rather than by assuming projective determinacy or large cardinals.

The forcing construction is based on the branch-versus-specialize coding matrix of Fischer, Friedman and Zdomskyy. Harrington's construction of long projective wellorders showed that strong projective definability can be compatible with a large continuum [5]. Fischer, Friedman and Zdomskyy later obtained a model of Martin's Axiom with $2^{\aleph_0} = \aleph_3$ and a lightface Δ_3^1 wellorder of the reals [3]. Their construction uses a matrix of Suslin trees and records information through the dichotomy between specializing a tree and adding a cofinal branch through it. We use this apparatus not only to recover the wellorder but also to make the truth values needed for uniformization projectively visible.

The main difficulty is not merely to insert enough codes. The iteration must simultaneously preserve the rigidity of the branch-versus-specialize coding, run enough c.c.c. forcing to obtain Martin's Axiom, keep the wellorder-coding blocks separate from the uniformization-coding blocks, and make minimal choices visible at the correct projective level. The last point is the delicate one. Choosing the least element of a nonempty projective section normally introduces a large initial-segment comparison, and a direct projective definition of this minimality assertion would have the wrong complexity. The construction therefore has to replace the minimality assertion by a coded acceptance predicate whose complexity is the same as that of the original section relation.

This is where the copying mechanism enters. We isolate a unary lightface Σ_3^1 predicate, denoted $\Psi_3(z)$, whose positive instances are exactly the reals deliberately inserted into

uniformization-reserved coding blocks. A projective formula is first put in normal form until the remaining matrix is Π_2^1 or Σ_2^1 . Once all real parameters and auxiliary witnesses have appeared, the truth value of each such matrix instance is stable by Shoenfield absoluteness. The forcing then copies this stable truth value into the Ψ_3 -pattern.

The power of the construction is that it does not only copy the truth value of a single Π_2^1 or Σ_2^1 formula. In the minimality part of the argument, the relevant assertion is a possibly infinite conjunction, or dually a possibly infinite disjunction, of such matrix tests, ranging over an initial segment of the final projective wellorder. The coding forcing, together with the copying idea, compresses this whole infinitary bookkeeping assertion into a single block-code statement. From the external point of view, the bookkeeping presents the required family of tests and the construction records whether the corresponding infinitary condition holds. From the internal point of view of the final model, this information is read by the ordinary projective predicate Ψ_3 . Thus an assertion which would be projectively too complicated if written out directly is replaced by a same-level copied acceptance predicate.

The compression used here is only a bookkeeping device in the construction. Long initial segments of the final wellorder, and the corresponding families of counter-witnesses needed to verify minimality, are represented externally by single reals when the iteration decides what to code next. These representatives are not part of the final projective formulas and need not themselves be projectively definable. The final definitions use only the projective wellorder, the predicate Ψ_3 , and the copied acceptance predicates produced by the iteration.

Unless explicitly stated otherwise, projective pointclasses are understood in the boldface sense. When formulas are enumerated, real parameters are included among the displayed real variables, or equivalently folded into the tuples being copied. Lightface definability is always indicated explicitly. The Fischer–Friedman–Zdomsky apparatus is recalled in Section 2. Section 3 records the external compression facts used by the bookkeeping. The branch-versus-specialize predicate is defined in Section 4, the final iteration in Section 5, and the copied projective predicates and uniformization argument in Sections 6 and 7. Section 8 records a cardinal-characteristic variant.

2 The Fischer–Friedman–Zdomsky apparatus

We recall the part of [3] needed in the sequel. All forcing is performed over L .

Fix a canonical sequence

$$\vec{S} = \langle S_\xi : 1 < \xi < \omega_3 \rangle$$

of stationary subsets of $\omega_2 \cap \text{cof}(\omega_1)$, Σ_1 -definable over L_{ω_3} with parameter ω_2 . Fix also a nicely definable almost-disjoint family

$$\vec{B} = \langle B_\xi : \xi < \omega_2 \rangle$$

of subsets of ω_1 , Σ_1 -definable over L_{ω_2} with parameter ω_1 . For $\xi < \omega_3$, let W_ξ denote the L -least subset of ω_2 coding ξ . We also fix the definable almost-disjoint family $\vec{C} = \langle C_{(\xi, \eta)} : \xi < \omega_1, \eta < \omega \cdot 3 \rangle$ of subsets of ω used at the final FFDZ real-coding stages.

Definition 2.1. A transitive model M of ZF^- is *suitable* if ω_2^M exists and

$$\omega_2^M = \omega_2^{L^M}.$$

Consequently $\omega_1^M = \omega_1^{L^M}$.

Throughout, ω_1 -trees are identified with subsets of ω_1 by the L -least bijection between $\omega^{<\omega_1}$ and ω_1 .

2.1 The preliminary mixed-support forcing

For $0 < \alpha < \omega_3$ and $n \in \omega$, let $\mathbb{K}_{\omega \cdot \alpha + n}^0$ be the standard countably closed Jech forcing adding a Suslin tree $T_{\omega \cdot \alpha + n}$ with countable conditions; see, for example, [15]. Put

$$\mathbb{K}_{0,\alpha} = \prod_{n \in \omega} \mathbb{K}_{\omega \cdot \alpha + n}^0$$

with full support.

In the extension by $\mathbb{K}_{0,\alpha}$, the tree $T_{\omega \cdot \alpha + n}$ is coded by killing stationarity of $S_{\omega_1 \cdot (\omega \cdot \alpha + n) + \gamma}$ exactly for $\gamma \in T_{\omega \cdot \alpha + n}$. More precisely, let

$$\mathbb{K}_{\alpha,n,\gamma}^1 = \begin{cases} \text{the forcing adding a club } C_{\omega_1 \cdot (\omega \cdot \alpha + n) + \gamma} \subseteq \omega_2 \setminus S_{\omega_1 \cdot (\omega \cdot \alpha + n) + \gamma}, & \gamma \in T_{\omega \cdot \alpha + n}, \\ \text{the trivial forcing,} & \gamma \notin T_{\omega \cdot \alpha + n}. \end{cases}$$

Let

$$\mathbb{K}_{1,\alpha,n} = \prod_{\gamma < \omega_1} \mathbb{K}_{\alpha,n,\gamma}^1, \quad \mathbb{K}_{1,\alpha} = \prod_{n \in \omega} \mathbb{K}_{1,\alpha,n},$$

again with full support.

For a set X of ordinals write

$$0(X) = \{\eta : 3\eta \in X\}, \quad I(X) = \{\eta : 3\eta + 1 \in X\}, \quad II(X) = \{\eta : 3\eta + 2 \in X\}.$$

Let $\chi : \omega_1 \times \omega_2 \rightarrow \omega_2$ be the fixed definable bijection. In $L^{\mathbb{K}_{0,\alpha} * \mathbb{K}_{1,\alpha}}$, let $D_{\omega \cdot \alpha + n} \subseteq \omega_2$ code

$$W_{\omega \cdot \alpha + n}, \quad W_{\omega \cdot \alpha}, \quad \langle C_{\omega_1 \cdot (\omega \cdot \alpha + n) + \gamma} : \gamma \in T_{\omega \cdot \alpha + n} \rangle$$

by requiring

$$0(D_{\omega \cdot \alpha + n}) = W_{\omega \cdot \alpha + n}, \quad I(D_{\omega \cdot \alpha + n}) = W_{\omega \cdot \alpha},$$

and

$$II(D_{\omega \cdot \alpha + n}) = \chi(\{\langle \gamma, \eta \rangle : \gamma \in T_{\omega \cdot \alpha + n}, \eta \in C_{\omega_1 \cdot (\omega \cdot \alpha + n) + \gamma}\}).$$

Choose $Z_{\omega \cdot \alpha + n} \subseteq \omega_2$ with even part $D_{\omega \cdot \alpha + n}$ and with the usual localization property: if $\beta < \omega_2$ is ω_2^M for a suitable M and $Z_{\omega \cdot \alpha + n} \cap \beta \in M$, then β lies in the relevant club of elementary submodels used in the Fischer–Friedman–Zdomsky construction.

Let $\psi_{\text{stat}}(\omega_1, \omega_2, Z, T, Z')$ be the Σ_1 formula saying that Z, Z' decode, through $\vec{B}$, canonical ordinal codes and clubs disjoint from the appropriate stationary sets, as in [3, Section 2]. Let $\varphi_{\text{loc}}(\omega_1, \omega_2, X, T, X')$ assert that, using $\vec{B}$, X and X' almost-disjointly code Z and Z' such that $\psi_{\text{stat}}(\omega_1, \omega_2, Z, T, Z')$ holds.

The third component $\mathbb{K}_{2,\alpha,n}$ adds $X_{\omega \cdot \alpha + n} \subseteq \omega_1$ almost-disjointly coding $Z_{\omega \cdot \alpha + n}$ by conditions $(s, s^*) \in [\omega_1]^{<\omega_1} \times [Z_{\omega \cdot \alpha + n}]^{<\omega_1}$, ordered by end-extension of s and the requirement that new points avoid B_ξ for $\xi \in s^*$. Let

$$\mathbb{K}_{2,\alpha} = \prod_{n \in \omega} \mathbb{K}_{2,\alpha,n}$$

with full support.

Finally, define the localization forcing

$$\mathbb{L}(X, T, X')$$

to consist of all functions $r : |r| \times 3 \rightarrow 2$ such that $|r|$ is a countable limit ordinal and:

- (1) for $\gamma < |r|$, $\gamma \in X$ iff $r(\gamma, 0) = 1$;

- (2) for $\gamma < |r|$, $\gamma \in X'$ iff $r(\gamma, 1) = 1$;
- (3) for every $\gamma \leq |r|$ and every suitable model M containing $r \upharpoonright (\gamma \times 3)$, the model M satisfies

$$\varphi_{\text{loc}}(\omega_1, \omega_2, X \cap \gamma, T \cap \gamma, X' \cap \gamma).$$

The order is end-extension: $q \leq p$ iff $|p| \leq |q|$ and $q \upharpoonright (|p| \times 3) = p$. This two-coordinate notation is used to avoid any closure assumption on a countable limit domain under ordinal maps of the form $\gamma \mapsto 3\gamma + i$. Put

$$\mathbb{K}_{3,\alpha,n} = \mathbb{L}(X_{\omega \cdot \alpha + n}, T_{\omega \cdot \alpha + n}, X_{\omega \cdot \alpha})$$

for $\alpha > 0$, with the obvious trivial definition at $\alpha = 0$, and set

$$\mathbb{K}_{3,\alpha} = \prod_{n \in \omega} \mathbb{K}_{3,\alpha,n}.$$

Let

$$\mathbb{K}_\alpha = \mathbb{K}_{0,\alpha} * \mathbb{K}_{1,\alpha} * \mathbb{K}_{2,\alpha} * \mathbb{K}_{3,\alpha}.$$

For $p \in \prod_{\alpha \in I} \mathbb{K}_\alpha$, define

$$\text{supp}_\omega(p) = \{\langle i, \alpha \rangle : i \in \{0, 2, 3\}, p_{i,\alpha} \neq \mathbf{1}\}$$

and

$$\text{supp}_{\omega_1}(p) = \{\langle 1, \alpha, n, \zeta \rangle : p_{1,\alpha,n,\zeta} \neq \mathbf{1}\}.$$

A condition has *mixed support* if $|\text{supp}_\omega(p)| \leq \omega$ and $|\text{supp}_{\omega_1}(p)| \leq \omega_1$. Let $\mathbb{P}_0$ be the mixed-support suborder of $\prod_{\alpha < \omega_3} \mathbb{K}_\alpha$.

Theorem 2.2 (Fischer–Friedman–Zdomskyy). *The forcing $\mathbb{P}_0$ is ω -distributive, has the ω_3 -chain condition, preserves cardinals, and preserves the relevant stationary sets in the following precise sense: if $\gamma \notin T_{\omega \cdot \alpha + n}$, then $S_{\omega_1 \cdot (\omega \cdot \alpha + n) + \gamma}$ remains stationary in $L^{\mathbb{P}_0}$.*

Proof. This is the preliminary-stage analysis of [3, Propositions 2.2, 2.3 and 2.6]. The distributivity proof uses the localization component $\mathbb{K}_3$ and the least-countable-elementary-submodel argument. The preservation of stationarity for $\gamma \notin T_{\omega \cdot \alpha + n}$ is obtained by the mixed-support genericity construction with the relation $\leq^*$. $\square$

2.2 The later c.c.c. iteration and the FFDZ wellorder

Over $L^{\mathbb{P}_0}$, Fischer–Friedman–Zdomskyy define a finite-support iteration

$$\langle \mathbb{P}_\alpha, \dot{\mathbb{Q}}_\beta : \alpha \leq \omega_3, \beta < \omega_3 \rangle$$

of c.c.c. posets. At successor stages, the bookkeeping forces all relevant instances of **MA** using names whose preliminary support is bounded below the current stage. At limit stages $\alpha = \omega \cdot \beta$, the iteration codes a pair of reals by specializing $T_{\omega \cdot \alpha + n}$ for n in a recursive transform $\Delta(x * y)$ and adding a branch through $T_{\omega \cdot \alpha + n}$ for $n \notin \Delta(x * y)$. A final almost-disjoint coding over ω produces a real R_α coding the resulting branch/specialization data.

Theorem 2.3 (Fischer–Friedman–Zdomskyy). *There is a finite-support c.c.c. iteration over $L^{\mathbb{P}_0}$ such that the final model satisfies*

$$\text{MA} + 2^{\aleph_0} = \aleph_3$$

and has a lightface Δ_3^1 wellorder of the reals.

Proof. This is [3, Theorem 2.9]. The key definition of the wellorder is given by the branch-versus-specialize pattern decoded in every countable suitable model containing the real added at the coding stage. $\square$

3 Bookkeeping compression and cardinal arithmetic

The uniformization construction needs to let the bookkeeping talk about long sequences of reals, for example families of counter-witnesses indexed by an initial segment of the final wellorder. The role of compression is purely external. We use cardinal arithmetic to choose, while defining the iteration, arbitrary representatives for such sequences. These representatives are not part of the final projective definitions and no projective definability of the representing maps is required.

We record the two elementary facts which are used repeatedly. The first is a bounded-support observation for names of long real sequences.

Lemma 3.1 (Bounded support for sequence names). *Let $\langle \mathbb{P}_\xi, \dot{\mathbb{Q}}_\xi : \xi \leq \omega_3 \rangle$ be a finite-support c.c.c. iteration whose iterands have size $< \aleph_3$. If $\alpha < \omega_3$, then every $\mathbb{P}_{\omega_3}$ -name for an α -sequence of reals is equivalent to a $\mathbb{P}_\beta$ -name for some $\beta < \omega_3$.*

Proof. Code an α -sequence of reals by a subset of $\alpha \times \omega$. For each coordinate $(\xi, n) \in \alpha \times \omega$, choose a countable maximal antichain deciding the corresponding bit. The union of the finite supports of all conditions appearing in these antichains has size at most $|\alpha| \cdot \aleph_0 \leq \aleph_2$. Since ω_3 is regular, this union is bounded in ω_3 . Hence the name is equivalent to a name over a proper initial segment of the iteration. $\square$

Lemma 3.2 (Final cardinal arithmetic). *For the forcing constructed below, the final model satisfies*

$$2^\omega = 2^{\omega_1} = 2^{\omega_2} = \aleph_3.$$

Consequently, for every $0 < \alpha < \omega_3$,

$$(2^\omega)^\alpha = 2^\omega.$$

Proof. The preliminary mixed-support forcing has size $\aleph_3$, preserves cardinals, and adds no reals. The later finite-support c.c.c. iteration has length ω_3 , all iterands have size $< \aleph_3$, and hence the full forcing has size $\aleph_3$. It adds cofinally many reals, so $2^\omega = \aleph_3$ in the final model.

Working in L , the number of names for subsets of ω_i , $i \in \{1, 2\}$, for a forcing of size $\aleph_3$ is at most

$$(\aleph_3)^{\omega_i} = \aleph_3,$$

using GCH. Since cardinals are preserved and $2^\omega = \aleph_3$, we get

$$2^{\omega_1} = 2^{\omega_2} = \aleph_3.$$

Finally, if $0 < \alpha < \omega_3$, then $|\alpha| \leq \omega_2$, and therefore

$$(2^\omega)^\alpha = \aleph_3^\alpha \leq \aleph_3^{\omega_2} = 2^{\omega_2} = \aleph_3 = 2^\omega.$$

The reverse inequality is trivial. $\square$

Corollary 3.3 (External compression of real sequences). *For every $0 < \alpha < \omega_3$, the bookkeeping may represent α -sequences of reals by single reals. The representing maps are chosen externally in the construction; they are not required to be definable in the final model and they are not mentioned by the final projective predicates.*

Proof. Lemma 3.2 gives $|(2^\omega)^\alpha| = 2^\omega$. Together with Lemma 3.1, this is enough to arrange that every relevant final sequence-name appears in the bookkeeping at some bounded stage. Once the construction decides to make information projective, it copies that information into the branch-versus-specialize pattern; the external representative itself carries no projective content. $\square$

4 A reusable branch-versus-specialize code

Fix a definable partition of the nonzero limit ordinals below ω_3 into two cofinal classes

$$\mathcal{L}_{\text{WO}} \quad \text{and} \quad \mathcal{L}_{\text{unif}}.$$

Blocks whose stage index belongs to $\mathcal{L}_{\text{WO}}$ are used for the FFDZ wellorder. Blocks whose stage index belongs to $\mathcal{L}_{\text{unif}}$ are reserved for the uniformization coding predicate below. This separation prevents a real coded only for the wellorder from becoming an accidental positive instance of Ψ_3 .

We now extract from the FFDZ wellorder proof a general coding predicate. This allows later uniformization stages to use the same robust coding matrix.

Fix once and for all a recursive injection

$$\Delta : 2^\omega \rightarrow 2^\omega$$

with uniformly recursive inverse on its range, and whose range consists of infinite coinfinite subsets of ω . In particular, the all-specialized pattern ω is not in the range of Δ . For example, one may let $\Delta(z)$ separate the bits of z from padding bits, as in the FFDZ coding of $x * y$.

Definition 4.1 (Block coding forcing). Let $\alpha \in \mathcal{L}_{\text{unif}}$ and let $z \in 2^\omega$ be a real in the current intermediate model V_α . The block base is $\omega \cdot \alpha$. Assume that each $T_{\omega \cdot \alpha + n}$ remains Suslin in V_α . The block coding forcing $\mathbb{Q}^{\text{blk}}(\alpha, z)$ is the two-step forcing

$$\mathbb{Q}^0(\alpha, z) * \dot{\mathbb{Q}}^1(\alpha),$$

where $\mathbb{Q}^0(\alpha, z)$ is the finite-support iteration

$$\langle \mathbb{Q}_n^0 : n \in \omega \rangle$$

such that $\mathbb{Q}_n^0$ specializes $T_{\omega \cdot \alpha + n}$ if $n \in \Delta(z)$, and forces with $T_{\omega \cdot \alpha + n}$ itself, thereby adding a cofinal branch, if $n \notin \Delta(z)$. In the extension by $\mathbb{Q}^0(\alpha, z)$, let $A_{\omega \cdot \alpha + n}$ denote the specializing function or branch added at coordinate n . The second forcing $\dot{\mathbb{Q}}^1(\alpha)$ is the usual FFDZ almost-disjoint coding over the fixed family $\vec{C}$, coding the sequences

$$\langle A_{\omega \cdot \alpha + n} : n \in \omega \rangle, \quad \langle Y_{\omega \cdot \alpha + n} : n \in \omega \rangle, \quad \langle T_{\omega \cdot \alpha + n} : n \in \omega \rangle$$

into a real R_α .

Definition 4.2 (Default block action). If $\alpha \in \mathcal{L}_{\text{unif}}$ and the block $\omega \cdot \alpha$ is not used to code a real, the default action at α is the analogous FFDZ block forcing which specializes every tree $T_{\omega \cdot \alpha + n}$, $n \in \omega$, and then almost-disjointly codes the resulting sequence of specializing functions together with the corresponding Y - and T -sequences into a real. Since the resulting pattern is the all-specialized pattern ω , and $\omega \notin \text{ran}(\Delta)$, a default block action cannot witness $\Psi_3(z)$ for any real z .

Lemma 4.3. *If $\langle T_{\omega \cdot \alpha + n} : n \in \omega \rangle$ is a sequence of Suslin trees in V_α , then both $\mathbb{Q}^{\text{blk}}(\alpha, z)$ and the default block action are c.c.c.*

Proof. This is the standard FFDZ mutual-preservation argument for the branch-versus-specialize matrix. Finite products and finite-support iterations of the relevant tree forcings and specialization forcings are c.c.c., and unused trees remain Suslin after such block actions. The final almost-disjoint coding over ω is c.c.c.; hence the two-step block forcing is c.c.c. The default action is the special case in which every coordinate uses the specialization forcing. $\square$

Definition 4.4 (The base predicate). For a real z , let $\Psi_3(z)$ be the following assertion:

there is a real R such that for every countable suitable model M with $R, z \in M$, there is an ordinal $\beta \in M$ such that M regards β as an element of the fixed class $\mathcal{L}_{\text{unif}}$ and the corresponding block base is

$$\tilde{\alpha} = \omega \cdot \beta$$

such that for every $n \in \omega$ the set

$$T_{\tilde{\alpha}, n}^M := \{\gamma < \omega_1^M : S_{\omega_1 \cdot (\tilde{\alpha} + n) + \gamma} \text{ is nonstationary in } M\}$$

is, under the canonical L -coding, an ω_1^M -tree, and

- (a) if $n \in \Delta(z)$, then M contains a specialization function for $T_{\tilde{\alpha}, n}^M$;
- (b) if $n \notin \Delta(z)$, then M contains a cofinal branch through $T_{\tilde{\alpha}, n}^M$.

No uniqueness of the witnessing block is required. The predicate is a unary assertion about z : in the final model it is intended to hold exactly for the reals deliberately placed into one of the uniformization-reserved coding blocks.

Lemma 4.5 (Complexity). *The relation $\Psi_3(z)$ is Σ_3^1 .*

Proof. The outer quantifier is the existential real R . The phrase “for every countable suitable model M containing R, z ” is a universal quantifier over reals coding well-founded countable structures. The remaining assertion is arithmetic in the code for M once the well-founded part is collapsed: suitability, the decoding of the nonstationarity pattern, and the existence in M of the relevant specializing functions or branches are all first-order assertions over the countable structure. The well-foundedness requirement contributes the usual coanalytic component, yielding an overall Σ_3^1 statement, exactly as in the FFDZ proof of the Δ_3^1 wellorder. $\square$

Lemma 4.6 (Unary exactness of the base code). *In the final model, for every real z ,*

$$\Psi_3(z)$$

holds if and only if z was deliberately placed by the iteration into one of the uniformization-reserved branch-versus-specialize coding blocks.

Proof. The forward direction for an intentionally used block is the argument of [3, Lemma 2.8], with $\Delta(z)$ replacing $\Delta(x * y)$. The real R_α added by the second step carries the data $A_{\omega \cdot \alpha + n}$, $Y_{\omega \cdot \alpha + n}$, and $T_{\omega \cdot \alpha + n}$ into every relevant countable suitable model. The localization forcing $\mathbb{K}_3$ ensures that the model decodes the stationary-kill pattern correctly.

Conversely, suppose $\Psi_3(w)$ holds. The witnessing block need not be unique and need not be recoverable from w . What matters is that any block witnessing the branch-versus-specialize pattern must be a uniformization-reserved block whose trees were prepared by the preliminary FFDZ forcing. The stationarity preservation part of Theorem 2.2 rules out phantom trees, and the later support restrictions prevent unused future blocks from acquiring an accidental branch/specialization pattern. Wellorder-reserved blocks are ignored by the definition of Ψ_3 , and auxiliary bookkeeping objects are never decoded by this predicate. Therefore the observed pattern must come from an actual uniformization block action of the iteration. Since each such action is assigned a single pattern $\Delta(z)$, the injectivity of Δ gives $w = z$ for the real deliberately coded at that block. $\square$

Lemma 4.7 (Jech trees remain Suslin in intermediate c.c.c. extensions). *Let $\mathbb{J}$ be the Jech forcing which adds one of the preliminary Suslin trees over the ground model. Although $\mathbb{J}$ need not remain literally σ -closed after an intervening c.c.c. extension, the tree it adds is still Suslin over the relevant intermediate model used in the FFDZ factorization.*

Proof. One must not argue that Jech's forcing remains σ -closed over the later intermediate c.c.c. universe. New countable descending sequences of old conditions may have appeared. Instead one uses the standard commutation/factorization of the independent pieces of the preliminary forcing. View the same final extension as obtained by first applying the Jech forcing over L , where the usual σ -closure argument shows that the generic tree is Suslin, and then applying the corresponding FFDZ tree-preserving c.c.c. forcing. The latter forcing preserves Suslinity of the newly added tree. Reordering the factors back to the original presentation shows that the tree is Suslin over the intermediate universe in which it is subsequently used. $\square$

Lemma 4.8 (No accidental future codes). *Let α be a limit block not yet used. In every intermediate model before stage α , the sequence $\langle T_{\omega \cdot \alpha + n} : n \in \omega \rangle$ remains a sequence of Suslin trees. Therefore no earlier c.c.c. forcing creates a branch/specialization pattern at block α .*

Proof. This follows from the FFDZ factorization argument, in particular [3, Claim 2.7], together with Lemma 4.7. The local-support machinery is the standard one from the FFDZ construction: the names used before the block have bounded preliminary support, and the quotient which prepares the future block can be commuted so that the Suslin tree is first added by Jech forcing over the ground model and only then exposed to the relevant tree-preserving c.c.c. forcing. Thus future blocks remain fresh until they are deliberately acted upon. $\square$

5 The final iteration

Let S_{MA} be a definable cofinal class of successor ordinals below ω_3 . We use the classes $\mathcal{L}_{\text{WO}}$ and $\mathcal{L}_{\text{unif}}$ from the previous section for the two kinds of limit-stage block coding. Let

$$F : \omega_3 \setminus \{0\} \rightarrow L_{\omega_3}$$

be a definable bookkeeping function such that every relevant name or tuple appears cofinally often in the appropriate part of the bookkeeping.

The iteration is a finite-support c.c.c. iteration over $L^{\mathbb{P}_0}$. During the recursive definition we also maintain an external forbidden set B_α of real codes. If $z \in B_\alpha$, then no later legitimate uniformization stage is allowed to use a block forcing which codes z . This is not a projective object in the final model; it is part of the metatheoretic definition of the iteration.

5.1 MA stages

If $\alpha \in S_{\text{MA}}$ and $F(\alpha)$ is a $\mathbb{P}_\alpha$ -name for a c.c.c. poset together with a list of $< \aleph_3$ many dense sets, and this name is supported below α in the preliminary coordinate, then let $\dot{Q}_\alpha$ be that name. Otherwise let $\dot{Q}_\alpha$ be trivial.

Lemma 5.1 (Bounded support for MA names). *Every final name for a c.c.c. poset of size $< \aleph_3$, together with $< \aleph_3$ many dense sets, is equivalent to a name appearing at some stage $\alpha \in S_{\text{MA}}$ and satisfying the strict support restriction.*

Proof. Finite-support c.c.c. iterations have nice names for sets of ordinals and for reals. Since ω_3 is regular and each name under consideration has hereditary size $< \omega_3$, the union of the preliminary supports appearing in a nice name is bounded in ω_3 . The bookkeeping function lists the bounded version of the name cofinally often after that bound. Thus the stage can force the corresponding instance of MA without touching future preliminary blocks. $\square$

5.2 Wellorder stages

If $\alpha \in \mathcal{L}_{\text{WO}}$, perform exactly the FFDZ wellorder coding stage: if $F(\alpha)$ is a pair of canonical names for reals in V_α , code the ordered pair according to the canonical stage-of-appearance wellorder; otherwise specialize all trees in the block. The definition of the final wellorder is the FFDZ definition with the additional requirement that the witnessing block index belongs to $\mathcal{L}_{\text{WO}}$. Since $\mathcal{L}_{\text{WO}}$ is fixed and definable, this does not change the projective complexity. It ensures that the uniformization-reserved blocks are invisible to the wellorder predicate.

5.3 Uniformization stages

If $\alpha \in \mathcal{L}_{\text{unif}}$, the bookkeeping object $F(\alpha)$ is interpreted as a copying request. Such a request consists of a real code z and a finite tuple of real parameters to be inserted into one of the remaining Π_2^1 or Σ_2^1 matrices described in Section 6. The request is evaluated in the current universe V_α .

If the request has value “code”, all involved names satisfy the strict preliminary support restriction, the block α is fresh, and $z \notin B_\alpha$, then we force with $\mathbb{Q}^{\text{blk}}(\alpha, z)$ and keep the forbidden set unchanged. If the request has value “do not code”, we put z into $B_{\alpha+1}$ and use the default block action at α . If the request is illegitimate or $z \in B_\alpha$, we also use the default block action. At limit stages the forbidden set is the union of the previous forbidden sets. Thus every uniformization-reserved block is deliberately assigned either a $\Delta(z)$ -pattern for a unique real z or the default all-specialized pattern outside the range of Δ .

The important point is that the truth value used to decide between “code” and “do not code” is a truth value of a fixed Π_2^1 or Σ_2^1 formula with all real parameters already present. By Shoenfield absoluteness this value is the same in all later forcing extensions produced by the iteration.

Lemma 5.2. *The final iteration is c.c.c., preserves cardinals, forces $2^{\aleph_0} = \aleph_3$, and satisfies MA.*

Proof. Each stage is c.c.c.: this is clear at the trivial stages, follows from the definition at the MA stages, follows from the FFDZ argument at the wellorder stages, and follows from Lemma 4.3 at the uniformization stages. A finite-support iteration of c.c.c. posets is c.c.c. The length is ω_3 , and the bookkeeping supplies cofinally many nontrivial real-adding stages, so the continuum is $\aleph_3$. Lemma 5.1 gives full MA. $\square$

6 Derived projective predicates and the copying mechanism

Let $\langle \theta_e : e \in \omega \rangle$ be a recursive enumeration of projective formulas in normal form. We use a fixed recursive coding of finite tuples of reals by reals and write

$$\langle e, x, y, a_0, \dots, a_k \rangle$$

for such a code. The base predicate is Ψ_3 ; the symbols Φ_e^n are reserved for the derived copied predicates.

The following absoluteness observation is the engine of the copying construction.

Lemma 6.1 (Absolute matrix values). *Let M be one of the intermediate universes of the construction, let N be a later set-forcing extension arising from the iteration, and let $\vec{a} \in M$ be a finite tuple of reals. If $\psi(\vec{a})$ is Π_2^1 or Σ_2^1 , then $M \models \psi(\vec{a})$ if and only if $N \models \psi(\vec{a})$.*

Proof. This is Shoenfield absoluteness, applied after all real parameters in $\vec{a}$ have been fixed. In the applications below, N is a later forcing extension of M arising in the iteration. $\square$

Remark 6.2 (What is copied). The construction does not try to decide a high projective formula by a projective test at a later stage. Instead, the high projective formula has already been decomposed until only a Π_2^1 or Σ_2^1 matrix remains. The bookkeeping presents concrete tuples of reals to be inserted into this matrix. By Lemma 6.1, each such matrix value is fixed once and for all. The iteration then records that stable truth value by either placing the associated tuple into the distinguished Ψ_3 predicate or by forbidding that tuple from ever being so coded.

For odd levels, write a Σ_{2m+1}^1 formula in the form

$$\theta_e(x, y) \equiv \exists a_0 \forall a_1 \exists a_2 \cdots \exists a_{2m-2} \psi_e(x, y, a_0, \dots, a_{2m-2}),$$

where $\psi_e \in \Pi_2^1$. The copied predicate is

$$\Phi_e^{2m+1}(x, y) \iff \exists a_0 \forall a_1 \exists a_2 \cdots \exists a_{2m-2} \Psi_3(\langle e, x, y, a_0, \dots, a_{2m-2} \rangle).$$

For even levels, write a Σ_{2m}^1 formula in the form

$$\theta_e(x, y) \equiv \exists a_0 \forall a_1 \exists a_2 \cdots \forall a_{2m-3} \psi_e(x, y, a_0, \dots, a_{2m-3}),$$

where $\psi_e \in \Sigma_2^1$. The copied predicate is

$$\Phi_e^{2m}(x, y) \iff \exists a_0 \forall a_1 \exists a_2 \cdots \forall a_{2m-3} \neg \Psi_3(\langle e, x, y, a_0, \dots, a_{2m-3} \rangle).$$

Lemma 6.3 (Complexity of the copied predicates). *For every $n \geq 3$, Φ_e^n is a Σ_n^1 relation.*

Proof. This follows from Lemma 4.5 and the displayed alternating quantifier definitions. In the even case the negation of the Σ_3^1 base predicate is placed under the final universal polarity, yielding the stated Σ_{2m}^1 complexity. $\square$

Lemma 6.4 (Copy-correctness for the derived predicates). *In the final extension, for every $n \geq 3$, every index e for a Σ_n^1 formula in the displayed normal form, and all reals x, y ,*

$$\Phi_e^n(x, y) \iff \theta_e(x, y).$$

Proof. Consider the odd case. The terminal matrix ψ_e is Π_2^1 . Whenever the bookkeeping hands the construction a concrete tuple

$$(x, y, a_0, \dots, a_{2m-2}),$$

Lemma 6.1 implies that the truth value of $\psi_e(x, y, a_0, \dots, a_{2m-2})$ is absolute between the current universe and the final model. The uniformization/copying stage uses this truth value as an instruction: if the matrix value is true, the associated tuple is placed into a distinguished Ψ_3 block; if the matrix value is false, the corresponding real code is put into the forbidden set and is never coded later. Lemma 4.6 then says that the final Ψ_3 predicate records exactly these choices and no accidental choices.

The quantifier pattern outside the matrix is not changed. Therefore replacing ψ_e by the corresponding Ψ_3 assertion gives an equivalent Σ_{2m+1}^1 predicate in the final model. The even case is dual: the remaining matrix is Σ_2^1 , and the construction records the complementary truth value, which is why $\neg \Psi_3$ appears in the displayed definition of Φ_e^{2m} . $\square$

6.1 Copying initial segments and minimality

The previous lemma copies the truth of the original section relation. Uniformization requires a second copying step: the assertion that a candidate is the first successful one in the fixed wellorder. The naive formula

$$\Phi_e^n(x, y) \wedge \forall y' (y' <_G y \rightarrow \neg \Phi_e^n(x, y'))$$

is the correct conceptual target, but it is not the formula we use as the final definition, since the displayed universal minimality clause has the wrong apparent complexity. The iteration instead produces a Σ_n^1 copy-coded acceptance predicate.

For each real x , enumerate pairs (y, a_0) by the lexicographic order induced by the FFDZ wellorder $<_G$, with y as the primary coordinate and a_0 as the secondary coordinate. Thus the least accepted pair has, in particular, the least possible output y ; only after y is fixed do we choose the least leading witness a_0 .

Lemma 6.5 (Compression for copied initial segments). *Let $\alpha < \omega_3$. Every family of reals indexed by an ordinal below α , in particular every family of counter-witnesses attached to the $<_G$ -initial segment of a candidate pair, can be represented by a single real for the purposes of the bookkeeping.*

Proof. By Corollary 3.3, $|(2^\omega)^\alpha| = 2^\omega$ for every $\alpha < \omega_3$. The representing real is chosen externally in the construction. No definability of the representing map is needed, and the representing map is not mentioned in the final projective formula. $\square$

Lemma 6.6 (Copy-coded minimality). *For every $n \geq 3$ and every e coding a Σ_n^1 formula, the uniformization stages produce a Σ_n^1 predicate*

$$\text{Accept}_e^n(x, y)$$

such that, in the final model,

$$\text{Accept}_e^n(x, y)$$

holds if and only if $\theta_e(x, y)$ holds and y is the $<_G$ -least real in the vertical section $\{u : \theta_e(x, u)\}$.

Proof. We describe the odd case; the even case is dual. Suppose

$$\theta_e(x, y) \equiv \exists a_0 \forall a_1 \exists a_2 \cdots \exists a_{2m-2} \psi_e(x, y, a_0, \dots, a_{2m-2}),$$

with $\psi_e \in \Pi_2^1$. For a candidate pair (y, a_0) , failure of an earlier pair (y', a'_0) is witnessed by a suitable tuple of counter-witnesses inserted into the same fixed Π_2^1 matrix. Once those reals have been supplied by the bookkeeping, the truth values of all corresponding matrix instances are absolute by Lemma 6.1.

The set of earlier pairs has order type some $\alpha < \omega_3$. Lemma 6.5 allows the family of counter-witnesses for this initial segment to be represented by one real in the bookkeeping. The construction then copies the infinite conjunction/disjunction of the associated absolute matrix values into the Ψ_3 pattern. This is exactly the mechanism used in the earlier global Σ_{n+2}^1 -uniformization construction [11]: after the bookkeeping supplies the tuples, the construction translates the relevant infinite family of Π_2^1 or Σ_2^1 tests into coded/not-coded assertions.

Thus the syntactically problematic minimality clause is replaced, in the final model, by an equivalent block-code predicate of the same Σ_n^1 complexity as the original section relation. The use of the lexicographic order on (y, a_0) with y primary ensures that the accepted output is the least possible y , not merely the first pair according to an order which might prioritize a_0 . $\square$

7 Uniformization

Definition 7.1 (Uniformizing graph). For e coding a Σ_n^1 formula $\theta_e(x, y)$, define the final uniformizing graph by

$$U_e^n(x, y) \iff \text{Accept}_e^n(x, y).$$

The naive least-witness formula is only the conceptual target; the displayed definition uses the copy-coded acceptance predicate supplied by Lemma 6.6.

Theorem 7.2 (Main theorem). *There is a generic extension of L satisfying*

$$\text{MA} + 2^{\aleph_0} = \aleph_3$$

such that the reals have a lightface Δ_3^1 wellorder and, for every $n \geq 2$, every Σ_n^1 set of pairs of reals admits a Σ_n^1 uniformization.

Proof. Let G be generic for the iteration described above. Lemma 5.2 gives MA and $2^{\aleph_0} = \aleph_3$. The wellorder stages reproduce the proof of Theorem 2.3, with witnesses restricted to the definable class $\mathcal{L}_{\text{WO}}$, so $V = L[G]$ has a lightface Δ_3^1 wellorder $<_G$.

The case $n = 2$ follows in ZFC from the Kondo–Novikov theorem. Now fix $n \geq 3$ and a Σ_n^1 formula $\theta_e(x, y)$. If the x -section of θ_e is empty, the uniformizing graph has no value at x . If it is nonempty, Lemma 6.6 supplies a Σ_n^1 predicate $\text{Accept}_e^n(x, y)$ which holds exactly of the $<_G$ -least output in the section. Therefore $U_e^n(x, y) \iff \text{Accept}_e^n(x, y)$ is single-valued, has domain equal to the projection of the original set, and is contained in the original set. Hence U_e^n is a Σ_n^1 uniformization. $\square$

8 A cardinal-characteristic variant

We briefly record that the construction can be combined with the cardinal-characteristic bookkeeping of Fischer–Friedman–Zdomsky. This gives a variant of the main theorem in which full Martin’s Axiom is replaced by a prescribed pattern of cardinal characteristics, using the cardinal-characteristic bookkeeping of Fischer–Friedman–Zdomsky and Brendle–Fischer [3, 2].

Theorem 8.1. *There is a generic extension in which $2^{\aleph_0} = \aleph_3$, the Σ_n^1 -uniformization property holds for every $n \geq 2$, and*

$$\mathfrak{p} = \mathfrak{b} = \aleph_2 < \mathfrak{a} = \mathfrak{s} = \mathfrak{c} = \aleph_3.$$

Proof. We use the same preliminary FFDZ coding apparatus and the same uniformization bookkeeping as in the proof of the main theorem. The only change is in the final successor-stage bookkeeping. Instead of arranging full Martin’s Axiom, we follow the cardinal-characteristic bookkeeping from Fischer–Friedman–Zdomsky [3].

First, below ω_2 , we add a $<^*$ -increasing unbounded scale

$$H = \langle h_\xi : \xi < \omega_2 \rangle$$

by the usual Hechler stages. This gives $\mathfrak{b} \leq \aleph_2$ in the final model, provided the later iteration preserves the unboundedness of H .

Above ω_2 , the bookkeeping is divided into three cofinal classes of successor stages. On one class we force with the Fischer–Steprans posets, as used in [2, 3], which preserve the unboundedness of H and add a real not split by the old reals. On a second class, when the bookkeeping presents a name for a maximal almost disjoint family of size at most $\aleph_2$, we

force with Brendle’s poset [2], which preserves the unboundedness of H and destroys the maximality of that family. On the third class we force with the σ -centered posets of size at most $\aleph_1$.

The additional stages used for the projective wellorder and for the Σ_n^1 -uniformization construction are precisely of the same FFDZ coding type as before. They are c.c.c. of size at most $\aleph_1$, or belong to the preliminary closed/distributive coding apparatus, and hence do not interfere with the preservation of the scale H . The compression maps used for the uniformization bookkeeping are external and do not add forcing stages. Thus the preservation argument from Fischer–Friedman–Zdomskyy goes through unchanged.

The final model has $2^{\aleph_0} = \aleph_3$. Since the iteration includes all relevant σ -centered posets of size at most $\aleph_1$, we obtain

$$MA_{<\omega_2}(\sigma\text{-centered}).$$

By Bell’s theorem [1], this implies $\mathfrak{p} = \aleph_2$. Since $\mathfrak{p} \leq \mathfrak{b}$ and the scale H remains unbounded of size $\aleph_2$, we get

$$\mathfrak{p} = \mathfrak{b} = \aleph_2.$$

The cofinally many Fischer–Steprans stages imply that no family of size $\leq \aleph_2$ splits all infinite subsets of ω , hence

$$\mathfrak{s} = \aleph_3.$$

Similarly, the cofinally many Brendle stages destroy every name for a maximal almost disjoint family of size at most $\aleph_2$, and therefore

$$\mathfrak{a} = \aleph_3.$$

Finally, the uniformization argument is unchanged: every copying requirement still appears cofinally often in the bookkeeping, and the FFDZ coding predicate has the same exactness and preservation properties as in the proof of the main theorem. Hence the Σ_n^1 -uniformization property holds for every $n \geq 3$. The case $n = 2$ holds in ZFC by the Kondo–Novikov theorem. Thus the final model satisfies the desired cardinal-characteristic pattern together with global Σ_n^1 -uniformization for all $n \geq 2$. $\square$

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